Data Wrangling

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Seemless data wrangeling

The tidyverse is a bunch of packages and functions written by the same folks that manage Rstudio. The tidyverse builds upon base R to allow for easier use of large datasets.

If you do not have the tidyverse installed please install it and load it `install.pakcages("tidyverse")`

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

When you load in the tidyverse notice that it loads in 8 different packages. You will not have to load these individually.

More info and cheet sheets can be found here: https://tidyr.tidyverse.org/index.html

Lets demonstrate some of the most useful functionality of some tidyverse functions. **Note** that this tutorial does not cover everything and **self-learning** may be required for specific functionality. We will cover the following:

- Data wrangling & manipulation
  - mutate()
  - select()
  - filter()
  - the pipe %>%
  - summarise()
  - group_by()
  - joining
  - pivotting
  - Integration with plotting

For this tutorial I will use the same microbiome dataset we used in previous classes for `ggplot`, except this dataset includes all fungi in the Phylum Ascomycota, not just one family.

```r
microbiome.fungi <- read.csv("07_ggplot/Bull_richness.csv")
str(microbiome.fungi)
```

```
## 'data.frame':    287 obs. of  16 variables:
##  $ SampleID       : chr  "Corn2017LeafObjective2Collection1T1R1CAH2" "Corn2017LeafObjective2Collectio
##  $ Crop           : chr  "Corn" "Corn" "Corn" "Corn" ...
##  $ Objective      : chr  "Objective 2" "Objective 2" "Objective 2" "Objective 2" ...
##  $ Collection     : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Compartment    : chr  "Leaf" "Leaf" "Leaf" "Leaf" ...
##  $ DateSampled    : chr  "6/26/17" "6/26/17" "6/26/17" "6/26/17" ...
##  $ GrowthStage    : chr  "V6" "V6" "V6" "V6" ...
##  $ Treatment      : chr  "Conv." "Conv." "Conv." "Conv." ...
##  $ Rep            : chr  "R1" "R1" "R1" "R1" ...
##  $ Sample         : chr  "A" "B" "C" "A" ...
##  $ Fungicide      : chr  "C" "C" "C" "F" ...
##  $ Target_organism: chr  "Fungi" "Fungi" "Fungi" "Fungi" ...
##  $ Location       : chr  "Kellogg Biological Station" "Kellogg Biological Station" "Kellogg Biologica
##  $ Experiment     : chr  "LTER" "LTER" "LTER" "LTER" ...
##  $ Year           : int  2017 2017 2017 2017 2017 2017 2017 2017 2017 2017 ...
##  $ richness       : int  9 6 5 7 4 2 3 8 4 4 ...
```

Lets start exploring the data.

Sometimes you may want to only select certain columns of your data. This is accomplished with the select function..

```r
microbiome.fungi2 <- select(microbiome.fungi, SampleID, Crop, Compartment:Fungicide, richness)
str(microbiome.fungi2)
```

`select()`

```
## 'data.frame':    287 obs. of  10 variables:
##  $ SampleID   : chr  "Corn2017LeafObjective2Collection1T1R1CAH2" "Corn2017LeafObjective2Collection1T
##  $ Crop       : chr  "Corn" "Corn" "Corn" "Corn" ...
##  $ Compartment: chr  "Leaf" "Leaf" "Leaf" "Leaf" ...
##  $ DateSampled: chr  "6/26/17" "6/26/17" "6/26/17" "6/26/17" ...
##  $ GrowthStage: chr  "V6" "V6" "V6" "V6" ...
##  $ Treatment  : chr  "Conv." "Conv." "Conv." "Conv." ...
##  $ Rep        : chr  "R1" "R1" "R1" "R1" ...
##  $ Sample     : chr  "A" "B" "C" "A" ...
##  $ Fungicide  : chr  "C" "C" "C" "F" ...
##  $ richness   : int  9 6 5 7 4 2 3 8 4 4 ...
```

Note the syntax. I listed all the column headers I wanted, but at the end I used `Compartment:Fungicide` because all these were in order and I can use the column to select them all.

`filter()` This is a convenient function that allows you to subset data fairly easily. However, I'm not sure how its much easier than other subset function

```r
head(filter(microbiome.fungi2, Treatment == "Conv."))
```

```
##                                   SampleID Crop Compartment DateSampled
## 1 Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2 Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3 Corn2017LeafObjective2Collection1T1R1CCB3 Corn        Leaf     6/26/17
## 4 Corn2017LeafObjective2Collection1T1R1FAC3 Corn        Leaf     6/26/17
## 5 Corn2017LeafObjective2Collection1T1R1FBD3 Corn        Leaf     6/26/17
## 6 Corn2017LeafObjective2Collection1T1R1FCE3 Corn        Leaf     6/26/17
##   GrowthStage Treatment Rep Sample Fungicide richness
## 1          V6     Conv.  R1      A         C        9
## 2          V6     Conv.  R1      B         C        6
## 3          V6     Conv.  R1      C         C        5
## 4          V6     Conv.  R1      A         F        7
## 5          V6     Conv.  R1      B         F        4
## 6          V6     Conv.  R1      C         F        2
```

```r
# A more complex using &
head(filter(microbiome.fungi2, Treatment == "Conv." & Fungicide == "C"))
```

```
##                                   SampleID Crop Compartment DateSampled
## 1 Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2 Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3 Corn2017LeafObjective2Collection1T1R1CCB3 Corn        Leaf     6/26/17
## 4 Corn2017LeafObjective2Collection1T1R2CAF3 Corn        Leaf     6/26/17
## 5 Corn2017LeafObjective2Collection1T1R2CBG3 Corn        Leaf     6/26/17
## 6 Corn2017LeafObjective2Collection1T1R2CCH3 Corn        Leaf     6/26/17
##   GrowthStage Treatment Rep Sample Fungicide richness
## 1          V6     Conv.  R1      A         C        9
## 2          V6     Conv.  R1      B         C        6
## 3          V6     Conv.  R1      C         C        5
## 4          V6     Conv.  R2      A         C        3
## 5          V6     Conv.  R2      B         C        8
## 6          V6     Conv.  R2      C         C        4
```

```r
# Another more complex example using or |
head(filter(microbiome.fungi2, Sample == "A" | Sample == "B")) # samples A or B
```

```
##                                   SampleID Crop Compartment DateSampled
## 1 Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2 Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3 Corn2017LeafObjective2Collection1T1R1FAC3 Corn        Leaf     6/26/17
## 4 Corn2017LeafObjective2Collection1T1R1FBD3 Corn        Leaf     6/26/17
## 5 Corn2017LeafObjective2Collection1T1R2CAF3 Corn        Leaf     6/26/17
## 6 Corn2017LeafObjective2Collection1T1R2CBG3 Corn        Leaf     6/26/17
##   GrowthStage Treatment Rep Sample Fungicide richness
## 1          V6     Conv.  R1      A         C        9
## 2          V6     Conv.  R1      B         C        6
## 3          V6     Conv.  R1      A         F        7
## 4          V6     Conv.  R1      B         F        4
## 5          V6     Conv.  R2      A         C        3
## 6          V6     Conv.  R2      B         C        8
```

mutate() Mutate allows us to quickly create new columns.

```
microbiome.fungi2$logRich <- log(microbiome.fungi2$richness)
# Create a new column called logRich
head(mutate(microbiome.fungi2, logRich = log(richness)))
```

```
##                                   SampleID Crop Compartment DateSampled
## 1 Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2 Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3 Corn2017LeafObjective2Collection1T1R1CCB3 Corn        Leaf     6/26/17
## 4 Corn2017LeafObjective2Collection1T1R1FAC3 Corn        Leaf     6/26/17
## 5 Corn2017LeafObjective2Collection1T1R1FBD3 Corn        Leaf     6/26/17
## 6 Corn2017LeafObjective2Collection1T1R1FCE3 Corn        Leaf     6/26/17
##   GrowthStage Treatment Rep Sample Fungicide richness   logRich
## 1          V6     Conv.  R1      A         C        9 2.1972246
## 2          V6     Conv.  R1      B         C        6 1.7917595
## 3          V6     Conv.  R1      C         C        5 1.6094379
## 4          V6     Conv.  R1      A         F        7 1.9459101
## 5          V6     Conv.  R1      B         F        4 1.3862944
## 6          V6     Conv.  R1      C         F        2 0.6931472
```

```
# Creating a new column which combines Crop and Treatment
head(mutate(microbiome.fungi2, Crop_Treatment = paste(Crop, Treatment)))
```

```
##                                   SampleID Crop Compartment DateSampled
## 1 Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2 Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3 Corn2017LeafObjective2Collection1T1R1CCB3 Corn        Leaf     6/26/17
## 4 Corn2017LeafObjective2Collection1T1R1FAC3 Corn        Leaf     6/26/17
## 5 Corn2017LeafObjective2Collection1T1R1FBD3 Corn        Leaf     6/26/17
## 6 Corn2017LeafObjective2Collection1T1R1FCE3 Corn        Leaf     6/26/17
##   GrowthStage Treatment Rep Sample Fungicide richness   logRich Crop_Treatment
## 1          V6     Conv.  R1      A         C        9 2.1972246     Corn Conv.
## 2          V6     Conv.  R1      B         C        6 1.7917595     Corn Conv.
## 3          V6     Conv.  R1      C         C        5 1.6094379     Corn Conv.
## 4          V6     Conv.  R1      A         F        7 1.9459101     Corn Conv.
## 5          V6     Conv.  R1      B         F        4 1.3862944     Corn Conv.
## 6          V6     Conv.  R1      C         F        2 0.6931472     Corn Conv.
```

the pipe %>% We can also combine multiple functions together. This is an important concept and is very powerful. It helps to try to visualize what you are doing to your data at each step. Basically the pipe allows you to input the previous data from the right side of the pipe into the function on the left (or next line).

In

We will combine all previous steps into one large string of functions.

When we do this we do not have to specify the data in each function. The data from the previous step is transfered to the next step.

```
select(microbiome.fungi, SampleID, Crop, Compartment:Fungicide, richness) # before
```

```
##                                       SampleID Crop Compartment DateSampled
```

```
## 1    Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2    Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3    Corn2017LeafObjective2Collection1T1R1CCB3 Corn        Leaf     6/26/17
## 4    Corn2017LeafObjective2Collection1T1R1FAC3 Corn        Leaf     6/26/17
## 5    Corn2017LeafObjective2Collection1T1R1FBD3 Corn        Leaf     6/26/17
## 6    Corn2017LeafObjective2Collection1T1R1FCE3 Corn        Leaf     6/26/17
## 7    Corn2017LeafObjective2Collection1T1R2CAF3 Corn        Leaf     6/26/17
## 8    Corn2017LeafObjective2Collection1T1R2CBG3 Corn        Leaf     6/26/17
## 9    Corn2017LeafObjective2Collection1T1R2CCH3 Corn        Leaf     6/26/17
## 10   Corn2017LeafObjective2Collection1T1R2FAA4 Corn        Leaf     6/26/17
## 11   Corn2017LeafObjective2Collection1T1R2FBB4 Corn        Leaf     6/26/17
## 12   Corn2017LeafObjective2Collection1T1R2FCC4 Corn        Leaf     6/26/17
## 13   Corn2017LeafObjective2Collection1T1R5CAD4 Corn        Leaf     6/26/17
## 14   Corn2017LeafObjective2Collection1T1R5CBE4 Corn        Leaf     6/26/17
## 15   Corn2017LeafObjective2Collection1T1R5CCF4 Corn        Leaf     6/26/17
## 16   Corn2017LeafObjective2Collection1T1R5FAG4 Corn        Leaf     6/26/17
## 17   Corn2017LeafObjective2Collection1T1R5FBH4 Corn        Leaf     6/26/17
## 18   Corn2017LeafObjective2Collection1T1R5FCA5 Corn        Leaf     6/26/17
## 19   Corn2017LeafObjective2Collection1T1R6CAB5 Corn        Leaf     6/26/17
## 20   Corn2017LeafObjective2Collection1T1R6CBC5 Corn        Leaf     6/26/17
## 21   Corn2017LeafObjective2Collection1T1R6CCD5 Corn        Leaf     6/26/17
## 22   Corn2017LeafObjective2Collection1T1R6FAE5 Corn        Leaf     6/26/17
## 23   Corn2017LeafObjective2Collection1T1R6FBF5 Corn        Leaf     6/26/17
## 24   Corn2017LeafObjective2Collection1T1R6FCG5 Corn        Leaf     6/26/17
## 25   Corn2017LeafObjective2Collection1T2R1CAH5 Corn        Leaf     6/26/17
## 26   Corn2017LeafObjective2Collection1T2R1CBA6 Corn        Leaf     6/26/17
## 27   Corn2017LeafObjective2Collection1T2R1CCB6 Corn        Leaf     6/26/17
## 28   Corn2017LeafObjective2Collection1T2R1FAC6 Corn        Leaf     6/26/17
## 29   Corn2017LeafObjective2Collection1T2R1FBE6 Corn        Leaf     6/26/17
## 30   Corn2017LeafObjective2Collection1T2R1FCD6 Corn        Leaf     6/26/17
## 31   Corn2017LeafObjective2Collection1T2R2CAF6 Corn        Leaf     6/26/17
## 32   Corn2017LeafObjective2Collection1T2R2CBG6 Corn        Leaf     6/26/17
## 33   Corn2017LeafObjective2Collection1T2R2CCH6 Corn        Leaf     6/26/17
## 34   Corn2017LeafObjective2Collection1T2R2FAA7 Corn        Leaf     6/26/17
## 35   Corn2017LeafObjective2Collection1T2R2FBB7 Corn        Leaf     6/26/17
## 36   Corn2017LeafObjective2Collection1T2R2FCC7 Corn        Leaf     6/26/17
## 37   Corn2017LeafObjective2Collection1T2R5CAD7 Corn        Leaf     6/26/17
## 38   Corn2017LeafObjective2Collection1T2R5CBE7 Corn        Leaf     6/26/17
## 39   Corn2017LeafObjective2Collection1T2R5CCF7 Corn        Leaf     6/26/17
## 40   Corn2017LeafObjective2Collection1T2R5FAG7 Corn        Leaf     6/26/17
## 41   Corn2017LeafObjective2Collection1T2R5FBH7 Corn        Leaf     6/26/17
## 42   Corn2017LeafObjective2Collection1T2R5FCA8 Corn        Leaf     6/26/17
## 43   Corn2017LeafObjective2Collection1T2R6CAB8 Corn        Leaf     6/26/17
## 44   Corn2017LeafObjective2Collection1T2R6CBC8 Corn        Leaf     6/26/17
## 45   Corn2017LeafObjective2Collection1T2R6CCD8 Corn        Leaf     6/26/17
## 46   Corn2017LeafObjective2Collection1T2R6FAE8 Corn        Leaf     6/26/17
## 47   Corn2017LeafObjective2Collection1T2R6FBF8 Corn        Leaf     6/26/17
## 48   Corn2017LeafObjective2Collection1T2R6FCG8 Corn        Leaf     6/26/17
## 49   Corn2017LeafObjective2Collection2T1R1CAC9 Corn        Leaf      7/5/17
## 50   Corn2017LeafObjective2Collection2T1R1CBD9 Corn        Leaf      7/5/17
## 51   Corn2017LeafObjective2Collection2T1R1CCE9 Corn        Leaf      7/5/17
## 52   Corn2017LeafObjective2Collection2T1R1FAH8 Corn        Leaf      7/5/17
## 53   Corn2017LeafObjective2Collection2T1R1FBA9 Corn        Leaf      7/5/17
## 54   Corn2017LeafObjective2Collection2T1R1FCB9 Corn        Leaf      7/5/17
```

```
## 55  Corn2017LeafObjective2Collection2T1R2CAA10 Corn        Leaf      7/5/17
## 56  Corn2017LeafObjective2Collection2T1R2CBC10 Corn        Leaf      7/5/17
## 57  Corn2017LeafObjective2Collection2T1R2CCD10 Corn        Leaf      7/5/17
## 58   Corn2017LeafObjective2Collection2T1R2FAF9 Corn        Leaf      7/5/17
## 59   Corn2017LeafObjective2Collection2T1R2FBG9 Corn        Leaf      7/5/17
## 60   Corn2017LeafObjective2Collection2T1R2FCH9 Corn        Leaf      7/5/17
## 61  Corn2017LeafObjective2Collection2T1R5CAH10 Corn        Leaf      7/5/17
## 62  Corn2017LeafObjective2Collection2T1R5CBA11 Corn        Leaf      7/5/17
## 63  Corn2017LeafObjective2Collection2T1R5CCC11 Corn        Leaf      7/5/17
## 64  Corn2017LeafObjective2Collection2T1R5FAE10 Corn        Leaf      7/5/17
## 65  Corn2017LeafObjective2Collection2T1R5FBF10 Corn        Leaf      7/5/17
## 66  Corn2017LeafObjective2Collection2T1R5FCG10 Corn        Leaf      7/5/17
## 67  Corn2017LeafObjective2Collection2T1R6CAB11 Corn        Leaf      7/5/17
## 68  Corn2017LeafObjective2Collection2T1R6CBG11 Corn        Leaf      7/5/17
## 69  Corn2017LeafObjective2Collection2T1R6CCH11 Corn        Leaf      7/5/17
## 70  Corn2017LeafObjective2Collection2T1R6FAD11 Corn        Leaf      7/5/17
## 71  Corn2017LeafObjective2Collection2T1R6FBE11 Corn        Leaf      7/5/17
## 72  Corn2017LeafObjective2Collection2T1R6FCF11 Corn        Leaf      7/5/17
## 73  Corn2017LeafObjective2Collection2T2R1CAD12 Corn        Leaf      7/5/17
## 74  Corn2017LeafObjective2Collection2T2R1CBE12 Corn        Leaf      7/5/17
## 75  Corn2017LeafObjective2Collection2T2R1CCF12 Corn        Leaf      7/5/17
## 76  Corn2017LeafObjective2Collection2T2R1FAA12 Corn        Leaf      7/5/17
## 77  Corn2017LeafObjective2Collection2T2R1FBB12 Corn        Leaf      7/5/17
## 78  Corn2017LeafObjective2Collection2T2R1FCC12 Corn        Leaf      7/5/17
## 79   Corn2017LeafObjective2Collection2T2R2CAB1 Corn        Leaf      7/5/17
## 80   Corn2017LeafObjective2Collection2T2R2CBC1 Corn        Leaf      7/5/17
## 81   Corn2017LeafObjective2Collection2T2R2CCD1 Corn        Leaf      7/5/17
## 82  Corn2017LeafObjective2Collection2T2R2FAG12 Corn        Leaf      7/5/17
## 83  Corn2017LeafObjective2Collection2T2R2FBH12 Corn        Leaf      7/5/17
## 84   Corn2017LeafObjective2Collection2T2R2FCA1 Corn        Leaf      7/5/17
## 85   Corn2017LeafObjective2Collection2T2R5CAH1 Corn        Leaf      7/5/17
## 86   Corn2017LeafObjective2Collection2T2R5CBA2 Corn        Leaf      7/5/17
## 87   Corn2017LeafObjective2Collection2T2R5CCB2 Corn        Leaf      7/5/17
## 88   Corn2017LeafObjective2Collection2T2R5FAE1 Corn        Leaf      7/5/17
## 89   Corn2017LeafObjective2Collection2T2R5FBF1 Corn        Leaf      7/5/17
## 90   Corn2017LeafObjective2Collection2T2R5FCG1 Corn        Leaf      7/5/17
## 91   Corn2017LeafObjective2Collection2T2R6CAF2 Corn        Leaf      7/5/17
## 92   Corn2017LeafObjective2Collection2T2R6CBG2 Corn        Leaf      7/5/17
## 93   Corn2017LeafObjective2Collection2T2R6CCH2 Corn        Leaf      7/5/17
## 94   Corn2017LeafObjective2Collection2T2R6FAC2 Corn        Leaf      7/5/17
## 95   Corn2017LeafObjective2Collection2T2R6FBD2 Corn        Leaf      7/5/17
## 96   Corn2017LeafObjective2Collection2T2R6FCE2 Corn        Leaf      7/5/17
## 97   Corn2017LeafObjective2Collection3T1R1CAA3 Corn        Leaf     7/31/17
## 98   Corn2017LeafObjective2Collection3T1R1CBB3 Corn        Leaf     7/31/17
## 99   Corn2017LeafObjective2Collection3T1R1CCC3 Corn        Leaf     7/31/17
## 100  Corn2017LeafObjective2Collection3T1R1FAD3 Corn        Leaf     7/31/17
## 101  Corn2017LeafObjective2Collection3T1R1FBE3 Corn        Leaf     7/31/17
## 102  Corn2017LeafObjective2Collection3T1R1FCF3 Corn        Leaf     7/31/17
## 103  Corn2017LeafObjective2Collection3T1R2CAG3 Corn        Leaf     7/31/17
## 104  Corn2017LeafObjective2Collection3T1R2CBH3 Corn        Leaf     7/31/17
## 105  Corn2017LeafObjective2Collection3T1R2CCA4 Corn        Leaf     7/31/17
## 106  Corn2017LeafObjective2Collection3T1R2FAB4 Corn        Leaf     7/31/17
## 107  Corn2017LeafObjective2Collection3T1R2FBC4 Corn        Leaf     7/31/17
## 108  Corn2017LeafObjective2Collection3T1R2FCD4 Corn        Leaf     7/31/17
```

```
## 109  Corn2017LeafObjective2Collection3T1R5CAE4 Corn        Leaf     7/31/17
## 110  Corn2017LeafObjective2Collection3T1R5CBF4 Corn        Leaf     7/31/17
## 111  Corn2017LeafObjective2Collection3T1R5CCG4 Corn        Leaf     7/31/17
## 112  Corn2017LeafObjective2Collection3T1R5FAH4 Corn        Leaf     7/31/17
## 113  Corn2017LeafObjective2Collection3T1R5FBA5 Corn        Leaf     7/31/17
## 114  Corn2017LeafObjective2Collection3T1R5FCB5 Corn        Leaf     7/31/17
## 115  Corn2017LeafObjective2Collection3T1R6CAC5 Corn        Leaf     7/31/17
## 116  Corn2017LeafObjective2Collection3T1R6CBD5 Corn        Leaf     7/31/17
## 117  Corn2017LeafObjective2Collection3T1R6CCE5 Corn        Leaf     7/31/17
## 118  Corn2017LeafObjective2Collection3T1R6FAF5 Corn        Leaf     7/31/17
## 119  Corn2017LeafObjective2Collection3T1R6FBG5 Corn        Leaf     7/31/17
## 120  Corn2017LeafObjective2Collection3T1R6FCH5 Corn        Leaf     7/31/17
## 121  Corn2017LeafObjective2Collection3T2R1CAB6 Corn        Leaf     7/31/17
## 122  Corn2017LeafObjective2Collection3T2R1CBC6 Corn        Leaf     7/31/17
## 123  Corn2017LeafObjective2Collection3T2R1CCD6 Corn        Leaf     7/31/17
## 124  Corn2017LeafObjective2Collection3T2R1FAE6 Corn        Leaf     7/31/17
## 125  Corn2017LeafObjective2Collection3T2R1FBF6 Corn        Leaf     7/31/17
## 126  Corn2017LeafObjective2Collection3T2R1FCG6 Corn        Leaf     7/31/17
## 127  Corn2017LeafObjective2Collection3T2R2CAH6 Corn        Leaf     7/31/17
## 128  Corn2017LeafObjective2Collection3T2R2CBA7 Corn        Leaf     7/31/17
## 129  Corn2017LeafObjective2Collection3T2R2CCB7 Corn        Leaf     7/31/17
## 130  Corn2017LeafObjective2Collection3T2R2FAC7 Corn        Leaf     7/31/17
## 131  Corn2017LeafObjective2Collection3T2R2FBD7 Corn        Leaf     7/31/17
## 132  Corn2017LeafObjective2Collection3T2R2FCE7 Corn        Leaf     7/31/17
## 133  Corn2017LeafObjective2Collection3T2R5CAF7 Corn        Leaf     7/31/17
## 134  Corn2017LeafObjective2Collection3T2R5CBG7 Corn        Leaf     7/31/17
## 135  Corn2017LeafObjective2Collection3T2R5CCH7 Corn        Leaf     7/31/17
## 136  Corn2017LeafObjective2Collection3T2R5FAA8 Corn        Leaf     7/31/17
## 137  Corn2017LeafObjective2Collection3T2R5FBB8 Corn        Leaf     7/31/17
## 138  Corn2017LeafObjective2Collection3T2R5FCC8 Corn        Leaf     7/31/17
## 139  Corn2017LeafObjective2Collection3T2R6CAD8 Corn        Leaf     7/31/17
## 140  Corn2017LeafObjective2Collection3T2R6CBE8 Corn        Leaf     7/31/17
## 141  Corn2017LeafObjective2Collection3T2R6CCF8 Corn        Leaf     7/31/17
## 142  Corn2017LeafObjective2Collection3T2R6FAG8 Corn        Leaf     7/31/17
## 143  Corn2017LeafObjective2Collection3T2R6FBH8 Corn        Leaf     7/31/17
## 144  Corn2017LeafObjective2Collection3T2R6FCA9 Corn        Leaf     7/31/17
## 145                                   T1R1AR4L  Soy        Leaf     8/16/18
## 146                                   T1R1BR4L  Soy        Leaf     8/16/18
## 147                                  T1R1CAR3L  Soy        Leaf      8/3/18
## 148                                  T1R1CAR6L  Soy        Leaf     8/27/18
## 149                                  T1R1CBR3L  Soy        Leaf      8/3/18
## 150                                  T1R1CBR6L  Soy        Leaf     8/27/18
## 151                                  T1R1CCR3L  Soy        Leaf      8/3/18
## 152                                  T1R1CCR6L  Soy        Leaf     8/27/18
## 153                                   T1R1CR4L  Soy        Leaf     8/16/18
## 154                                  T1R1FAR3L  Soy        Leaf      8/3/18
## 155                                  T1R1FAR4L  Soy        Leaf     8/16/18
## 156                                  T1R1FAR6L  Soy        Leaf     8/27/18
## 157                                  T1R1FBR3L  Soy        Leaf      8/3/18
## 158                                  T1R1FBR4L  Soy        Leaf     8/16/18
## 159                                  T1R1FBR6L  Soy        Leaf     8/27/18
## 160                                  T1R1FCR3L  Soy        Leaf      8/3/18
## 161                                  T1R1FCR4L  Soy        Leaf     8/16/18
## 162                                  T1R1FCR6L  Soy        Leaf     8/27/18
```

```
## 163                                    T1R2CAR3L  Soy        Leaf      8/3/18
## 164                                    T1R2CAR4L  Soy        Leaf     8/16/18
## 165                                    T1R2CAR6L  Soy        Leaf     8/27/18
## 166                                    T1R2CBR3L  Soy        Leaf      8/3/18
## 167                                    T1R2CBR4L  Soy        Leaf     8/16/18
## 168                                    T1R2CBR6L  Soy        Leaf     8/27/18
## 169                                    T1R2CCR3L  Soy        Leaf      8/3/18
## 170                                    T1R2CCR4L  Soy        Leaf     8/16/18
## 171                                    T1R2CCR6L  Soy        Leaf     8/27/18
## 172                                    T1R2FAR3L  Soy        Leaf      8/3/18
## 173                                    T1R2FAR4L  Soy        Leaf     8/16/18
## 174                                    T1R2FAR6L  Soy        Leaf     8/27/18
## 175                                    T1R2FBR3L  Soy        Leaf      8/3/18
## 176                                    T1R2FBR4L  Soy        Leaf     8/16/18
## 177                                    T1R2FBR6L  Soy        Leaf     8/27/18
## 178                                    T1R2FCR3L  Soy        Leaf      8/3/18
## 179                                    T1R2FCR4L  Soy        Leaf     8/16/18
## 180                                    T1R2FCR6L  Soy        Leaf     8/27/18
## 181                                    T1R5CAR3L  Soy        Leaf      8/3/18
## 182                                    T1R5CAR4L  Soy        Leaf     8/16/18
## 183                                    T1R5CAR6L  Soy        Leaf     8/27/18
## 184                                    T1R5CBR3L  Soy        Leaf      8/3/18
## 185                                    T1R5CBR4L  Soy        Leaf     8/16/18
## 186                                    T1R5CBR6L  Soy        Leaf     8/27/18
## 187                                    T1R5CCR3L  Soy        Leaf      8/3/18
## 188                                    T1R5CCR4L  Soy        Leaf     8/16/18
## 189                                    T1R5CCR6L  Soy        Leaf     8/27/18
## 190                                    T1R5FAR3L  Soy        Leaf      8/3/18
## 191                                    T1R5FAR4L  Soy        Leaf     8/16/18
## 192                                    T1R5FAR6L  Soy        Leaf     8/27/18
## 193                                    T1R5FBR3L  Soy        Leaf      8/3/18
## 194                                    T1R5FBR4L  Soy        Leaf     8/16/18
## 195                                    T1R5FBR6L  Soy        Leaf     8/27/18
## 196                                    T1R5FCR3L  Soy        Leaf      8/3/18
## 197                                    T1R5FCR4L  Soy        Leaf     8/16/18
## 198                                    T1R5FCR6L  Soy        Leaf     8/27/18
## 199                                    T1R6CAR3L  Soy        Leaf      8/3/18
## 200                                    T1R6CAR4L  Soy        Leaf     8/16/18
## 201                                    T1R6CAR6L  Soy        Leaf     8/27/18
## 202                                    T1R6CBR3L  Soy        Leaf      8/3/18
## 203                                    T1R6CBR4L  Soy        Leaf     8/16/18
## 204                                    T1R6CBR6L  Soy        Leaf     8/27/18
## 205                                    T1R6CCR3L  Soy        Leaf      8/3/18
## 206                                    T1R6CCR4L  Soy        Leaf     8/16/18
## 207                                    T1R6CCR6L  Soy        Leaf     8/27/18
## 208                                    T1R6FAR3L  Soy        Leaf      8/3/18
## 209                                    T1R6FAR4L  Soy        Leaf     8/16/18
## 210                                    T1R6FAR6L  Soy        Leaf     8/27/18
## 211                                    T1R6FBR3L  Soy        Leaf      8/3/18
## 212                                    T1R6FBR4L  Soy        Leaf     8/16/18
## 213                                    T1R6FBR6L  Soy        Leaf     8/27/18
## 214                                    T1R6FCR3L  Soy        Leaf      8/3/18
## 215                                    T1R6FCR4L  Soy        Leaf     8/16/18
## 216                                    T1R6FCR6L  Soy        Leaf     8/27/18
```

```
## 217                                    T2R1CAR3L  Soy        Leaf      8/3/18
## 218                                    T2R1CAR4L  Soy        Leaf     8/16/18
## 219                                    T2R1CAR6L  Soy        Leaf     8/27/18
## 220                                    T2R1CBR3L  Soy        Leaf      8/3/18
## 221                                    T2R1CBR4L  Soy        Leaf     8/16/18
## 222                                    T2R1CBR6L  Soy        Leaf     8/27/18
## 223                                    T2R1CCR3L  Soy        Leaf      8/3/18
## 224                                    T2R1CCR4L  Soy        Leaf     8/16/18
## 225                                    T2R1CCR6L  Soy        Leaf     8/27/18
## 226                                    T2R1FAR3L  Soy        Leaf      8/3/18
## 227                                    T2R1FAR4L  Soy        Leaf     8/16/18
## 228                                    T2R1FAR6L  Soy        Leaf     8/27/18
## 229                                    T2R1FBR3L  Soy        Leaf      8/3/18
## 230                                    T2R1FBR4L  Soy        Leaf     8/16/18
## 231                                    T2R1FBR6L  Soy        Leaf     8/27/18
## 232                                    T2R1FCR3L  Soy        Leaf      8/3/18
## 233                                    T2R1FCR4L  Soy        Leaf     8/16/18
## 234                                    T2R2CAR3L  Soy        Leaf      8/3/18
## 235                                    T2R2CAR4L  Soy        Leaf     8/16/18
## 236                                    T2R2CAR6L  Soy        Leaf     8/27/18
## 237                                    T2R2CBR3L  Soy        Leaf      8/3/18
## 238                                    T2R2CBR4L  Soy        Leaf     8/16/18
## 239                                    T2R2CBR6L  Soy        Leaf     8/27/18
## 240                                    T2R2CCR3L  Soy        Leaf      8/3/18
## 241                                    T2R2CCR4L  Soy        Leaf     8/16/18
## 242                                    T2R2CCR6L  Soy        Leaf     8/27/18
## 243                                    T2R2FAR3L  Soy        Leaf      8/3/18
## 244                                    T2R2FAR4L  Soy        Leaf     8/16/18
## 245                                    T2R2FAR6L  Soy        Leaf     8/27/18
## 246                                    T2R2FBR3L  Soy        Leaf      8/3/18
## 247                                    T2R2FBR4L  Soy        Leaf     8/16/18
## 248                                    T2R2FBR6L  Soy        Leaf     8/27/18
## 249                                    T2R2FCR3L  Soy        Leaf      8/3/18
## 250                                    T2R2FCR4L  Soy        Leaf     8/16/18
## 251                                    T2R2FCR6L  Soy        Leaf     8/27/18
## 252                                    T2R5CAR3L  Soy        Leaf      8/3/18
## 253                                    T2R5CAR4L  Soy        Leaf     8/16/18
## 254                                    T2R5CAR6L  Soy        Leaf     8/27/18
## 255                                    T2R5CBR3L  Soy        Leaf      8/3/18
## 256                                    T2R5CBR4L  Soy        Leaf     8/16/18
## 257                                    T2R5CBR6L  Soy        Leaf     8/27/18
## 258                                    T2R5CCR3L  Soy        Leaf      8/3/18
## 259                                    T2R5CCR4L  Soy        Leaf     8/16/18
## 260                                    T2R5CCR6L  Soy        Leaf     8/27/18
## 261                                    T2R5FAR3L  Soy        Leaf      8/3/18
## 262                                    T2R5FAR4L  Soy        Leaf     8/16/18
## 263                                    T2R5FAR6L  Soy        Leaf     8/27/18
## 264                                    T2R5FBR3L  Soy        Leaf      8/3/18
## 265                                    T2R5FBR4L  Soy        Leaf     8/16/18
## 266                                    T2R5FBR6L  Soy        Leaf     8/27/18
## 267                                    T2R5FCR3L  Soy        Leaf      8/3/18
## 268                                    T2R5FCR4L  Soy        Leaf     8/16/18
## 269                                    T2R5FCR6L  Soy        Leaf     8/27/18
## 270                                    T2R6CAR3L  Soy        Leaf      8/3/18
```

```
## 271                                  T2R6CAR4L  Soy        Leaf     8/16/18
## 272                                  T2R6CAR6L  Soy        Leaf     8/27/18
## 273                                  T2R6CBR3L  Soy        Leaf      8/3/18
## 274                                  T2R6CBR4L  Soy        Leaf     8/16/18
## 275                                  T2R6CBR6L  Soy        Leaf     8/27/18
## 276                                  T2R6CCR3L  Soy        Leaf      8/3/18
## 277                                  T2R6CCR4L  Soy        Leaf     8/16/18
## 278                                  T2R6CCR6L  Soy        Leaf     8/27/18
## 279                                  T2R6FAR3L  Soy        Leaf      8/3/18
## 280                                  T2R6FAR4L  Soy        Leaf     8/16/18
## 281                                  T2R6FAR6L  Soy        Leaf     8/27/18
## 282                                  T2R6FBR3L  Soy        Leaf      8/3/18
## 283                                  T2R6FBR4L  Soy        Leaf     8/16/18
## 284                                  T2R6FBR6L  Soy        Leaf     8/27/18
## 285                                  T2R6FCR3L  Soy        Leaf      8/3/18
## 286                                  T2R6FCR4L  Soy        Leaf     8/16/18
## 287                                  T2R6FCR6L  Soy        Leaf     8/27/18
##     GrowthStage Treatment Rep Sample Fungicide richness
## 1            V6     Conv.  R1      A         C        9
## 2            V6     Conv.  R1      B         C        6
## 3            V6     Conv.  R1      C         C        5
## 4            V6     Conv.  R1      A         F        7
## 5            V6     Conv.  R1      B         F        4
## 6            V6     Conv.  R1      C         F        2
## 7            V6     Conv.  R2      A         C        3
## 8            V6     Conv.  R2      B         C        8
## 9            V6     Conv.  R2      C         C        4
## 10           V6     Conv.  R2      A         F        4
## 11           V6     Conv.  R2      B         F        3
## 12           V6     Conv.  R2      C         F        5
## 13           V6     Conv.  R5      A         C        5
## 14           V6     Conv.  R5      B         C        4
## 15           V6     Conv.  R5      C         C       10
## 16           V6     Conv.  R5      A         F        4
## 17           V6     Conv.  R5      B         F        6
## 18           V6     Conv.  R5      C         F        6
## 19           V6     Conv.  R6      A         C        2
## 20           V6     Conv.  R6      B         C        1
## 21           V6     Conv.  R6      C         C        6
## 22           V6     Conv.  R6      A         F        1
## 23           V6     Conv.  R6      B         F        4
## 24           V6     Conv.  R6      C         F        2
## 25           V6   No-till  R1      A         C        6
## 26           V6   No-till  R1      B         C        3
## 27           V6   No-till  R1      C         C        4
## 28           V6   No-till  R1      A         F       15
## 29           V6   No-till  R1      B         F        5
## 30           V6   No-till  R1      C         F        9
## 31           V6   No-till  R2      A         C        5
## 32           V6   No-till  R2      B         C        5
## 33           V6   No-till  R2      C         C        6
## 34           V6   No-till  R2      A         F       15
## 35           V6   No-till  R2      B         F        3
## 36           V6   No-till  R2      C         F        2
```

```
## 37          V6   No-till  R5      A         C        9
## 38          V6   No-till  R5      B         C        9
## 39          V6   No-till  R5      C         C        3
## 40          V6   No-till  R5      A         F        8
## 41          V6   No-till  R5      B         F        1
## 42          V6   No-till  R5      C         F       12
## 43          V6   No-till  R6      A         C       11
## 44          V6   No-till  R6      B         C        7
## 45          V6   No-till  R6      C         C        7
## 46          V6   No-till  R6      A         F        6
## 47          V6   No-till  R6      B         F        6
## 48          V6   No-till  R6      C         F        3
## 49          V8     Conv.  R1      A         C       19
## 50          V8     Conv.  R1      B         C       11
## 51          V8     Conv.  R1      C         C       10
## 52          V8     Conv.  R1      A         F        4
## 53          V8     Conv.  R1      B         F        3
## 54          V8     Conv.  R1      C         F       10
## 55          V8     Conv.  R2      A         C       14
## 56          V8     Conv.  R2      B         C        7
## 57          V8     Conv.  R2      C         C       14
## 58          V8     Conv.  R2      A         F        3
## 59          V8     Conv.  R2      B         F        4
## 60          V8     Conv.  R2      C         F        2
## 61          V8     Conv.  R5      A         C        7
## 62          V8     Conv.  R5      B         C       17
## 63          V8     Conv.  R5      C         C       10
## 64          V8     Conv.  R5      A         F        4
## 65          V8     Conv.  R5      B         F        6
## 66          V8     Conv.  R5      C         F        3
## 67          V8     Conv.  R6      A         C       17
## 68          V8     Conv.  R6      B         C        5
## 69          V8     Conv.  R6      C         C       10
## 70          V8     Conv.  R6      A         F        2
## 71          V8     Conv.  R6      B         F        8
## 72          V8     Conv.  R6      C         F        6
## 73          V8   No-till  R1      A         C       10
## 74          V8   No-till  R1      B         C       19
## 75          V8   No-till  R1      C         C       19
## 76          V8   No-till  R1      A         F        4
## 77          V8   No-till  R1      B         F        7
## 78          V8   No-till  R1      C         F        6
## 79          V8   No-till  R2      A         C       18
## 80          V8   No-till  R2      B         C       11
## 81          V8   No-till  R2      C         C       11
## 82          V8   No-till  R2      A         F        6
## 83          V8   No-till  R2      B         F        8
## 84          V8   No-till  R2      C         F       10
## 85          V8   No-till  R5      A         C       17
## 86          V8   No-till  R5      B         C       13
## 87          V8   No-till  R5      C         C       15
## 88          V8   No-till  R5      A         F       11
## 89          V8   No-till  R5      B         F        8
## 90          V8   No-till  R5      C         F        6
```

```
## 91          V8   No-till  R6      A         C        6
## 92          V8   No-till  R6      B         C       13
## 93          V8   No-till  R6      C         C       21
## 94          V8   No-till  R6      A         F       10
## 95          V8   No-till  R6      B         F        7
## 96          V8   No-till  R6      C         F        6
## 97         V15     Conv.  R1      A         C       15
## 98         V15     Conv.  R1      B         C       16
## 99         V15     Conv.  R1      C         C       12
## 100        V15     Conv.  R1      A         F       15
## 101        V15     Conv.  R1      B         F       14
## 102        V15     Conv.  R1      C         F       10
## 103        V15     Conv.  R2      A         C       11
## 104        V15     Conv.  R2      B         C       12
## 105        V15     Conv.  R2      C         C       12
## 106        V15     Conv.  R2      A         F       17
## 107        V15     Conv.  R2      B         F       16
## 108        V15     Conv.  R2      C         F        8
## 109        V15     Conv.  R5      A         C       22
## 110        V15     Conv.  R5      B         C       26
## 111        V15     Conv.  R5      C         C       11
## 112        V15     Conv.  R5      A         F       23
## 113        V15     Conv.  R5      B         F       17
## 114        V15     Conv.  R5      C         F        5
## 115        V15     Conv.  R6      A         C       16
## 116        V15     Conv.  R6      B         C       10
## 117        V15     Conv.  R6      C         C       10
## 118        V15     Conv.  R6      A         F        8
## 119        V15     Conv.  R6      B         F       12
## 120        V15     Conv.  R6      C         F       17
## 121        V15   No-till  R1      A         C        8
## 122        V15   No-till  R1      B         C       14
## 123        V15   No-till  R1      C         C        7
## 124        V15   No-till  R1      A         F       15
## 125        V15   No-till  R1      B         F       11
## 126        V15   No-till  R1      C         F       19
## 127        V15   No-till  R2      A         C       15
## 128        V15   No-till  R2      B         C       13
## 129        V15   No-till  R2      C         C       16
## 130        V15   No-till  R2      A         F       20
## 131        V15   No-till  R2      B         F       32
## 132        V15   No-till  R2      C         F       31
## 133        V15   No-till  R5      A         C       25
## 134        V15   No-till  R5      B         C       18
## 135        V15   No-till  R5      C         C       20
## 136        V15   No-till  R5      A         F       33
## 137        V15   No-till  R5      B         F       33
## 138        V15   No-till  R5      C         F       14
## 139        V15   No-till  R6      A         C       12
## 140        V15   No-till  R6      B         C       22
## 141        V15   No-till  R6      C         C       21
## 142        V15   No-till  R6      A         F       36
## 143        V15   No-till  R6      B         F       18
## 144        V15   No-till  R6      C         F       29
```

```
## 145          R4     Conv.  R1      A          C        21
## 146          R4     Conv.  R1      B          C        12
## 147          R3     Conv.  R1      A          C        15
## 148          R6     Conv.  R1      A          C        22
## 149          R3     Conv.  R1      B          C        18
## 150          R6     Conv.  R1      B          C        20
## 151          R3     Conv.  R1      C          C        18
## 152          R6     Conv.  R1      C          C        21
## 153          R4     Conv.  R1      C          C        19
## 154          R3     Conv.  R1      A          F        15
## 155          R4     Conv.  R1      A          F        11
## 156          R6     Conv.  R1      A          F        10
## 157          R3     Conv.  R1      B          F        10
## 158          R4     Conv.  R1      B          F         7
## 159          R6     Conv.  R1      B          F         6
## 160          R3     Conv.  R1      C          F        16
## 161          R4     Conv.  R1      C          F        16
## 162          R6     Conv.  R1      C          F        15
## 163          R3     Conv.  R2      A          C         7
## 164          R4     Conv.  R2      A          C        28
## 165          R6     Conv.  R2      A          C        22
## 166          R3     Conv.  R2      B          C         9
## 167          R4     Conv.  R2      B          C        19
## 168          R6     Conv.  R2      B          C        15
## 169          R3     Conv.  R2      C          C        14
## 170          R4     Conv.  R2      C          C        21
## 171          R6     Conv.  R2      C          C        18
## 172          R3     Conv.  R2      A          F        13
## 173          R4     Conv.  R2      A          F        11
## 174          R6     Conv.  R2      A          F        12
## 175          R3     Conv.  R2      B          F        12
## 176          R4     Conv.  R2      B          F         8
## 177          R6     Conv.  R2      B          F         4
## 178          R3     Conv.  R2      C          F        15
## 179          R4     Conv.  R2      C          F         8
## 180          R6     Conv.  R2      C          F        10
## 181          R3     Conv.  R5      A          C        25
## 182          R4     Conv.  R5      A          C        18
## 183          R6     Conv.  R5      A          C        23
## 184          R3     Conv.  R5      B          C        18
## 185          R4     Conv.  R5      B          C        16
## 186          R6     Conv.  R5      B          C        22
## 187          R3     Conv.  R5      C          C        19
## 188          R4     Conv.  R5      C          C        14
## 189          R6     Conv.  R5      C          C        17
## 190          R3     Conv.  R5      A          F        22
## 191          R4     Conv.  R5      A          F         9
## 192          R6     Conv.  R5      A          F         7
## 193          R3     Conv.  R5      B          F        13
## 194          R4     Conv.  R5      B          F        14
## 195          R6     Conv.  R5      B          F         9
## 196          R3     Conv.  R5      C          F        23
## 197          R4     Conv.  R5      C          F         9
## 198          R6     Conv.  R5      C          F         6
```

```
## 199          R3     Conv.  R6      A         C       25
## 200          R4     Conv.  R6      A         C       22
## 201          R6     Conv.  R6      A         C       20
## 202          R3     Conv.  R6      B         C       27
## 203          R4     Conv.  R6      B         C       19
## 204          R6     Conv.  R6      B         C       15
## 205          R3     Conv.  R6      C         C       23
## 206          R4     Conv.  R6      C         C       22
## 207          R6     Conv.  R6      C         C       13
## 208          R3     Conv.  R6      A         F       18
## 209          R4     Conv.  R6      A         F       13
## 210          R6     Conv.  R6      A         F       17
## 211          R3     Conv.  R6      B         F       19
## 212          R4     Conv.  R6      B         F       10
## 213          R6     Conv.  R6      B         F       11
## 214          R3     Conv.  R6      C         F       17
## 215          R4     Conv.  R6      C         F        5
## 216          R6     Conv.  R6      C         F       16
## 217          R3   No-till  R1      A         C       20
## 218          R4   No-till  R1      A         C       25
## 219          R6   No-till  R1      A         C       18
## 220          R3   No-till  R1      B         C       21
## 221          R4   No-till  R1      B         C       17
## 222          R6   No-till  R1      B         C       15
## 223          R3   No-till  R1      C         C       22
## 224          R4   No-till  R1      C         C       27
## 225          R6   No-till  R1      C         C       20
## 226          R3   No-till  R1      A         F       25
## 227          R4   No-till  R1      A         F       11
## 228          R6   No-till  R1      A         F       12
## 229          R3   No-till  R1      B         F       39
## 230          R4   No-till  R1      B         F       12
## 231          R6   No-till  R1      B         F        9
## 232          R3   No-till  R1      C         F       19
## 233          R4   No-till  R1      C         F        7
## 234          R3   No-till  R2      A         C       18
## 235          R4   No-till  R2      A         C       24
## 236          R6   No-till  R2      A         C       16
## 237          R3   No-till  R2      B         C       18
## 238          R4   No-till  R2      B         C       17
## 239          R6   No-till  R2      B         C       16
## 240          R3   No-till  R2      C         C       14
## 241          R4   No-till  R2      C         C       22
## 242          R6   No-till  R2      C         C       25
## 243          R3   No-till  R2      A         F       22
## 244          R4   No-till  R2      A         F       10
## 245          R6   No-till  R2      A         F       13
## 246          R3   No-till  R2      B         F       13
## 247          R4   No-till  R2      B         F        8
## 248          R6   No-till  R2      B         F       13
## 249          R3   No-till  R2      C         F       15
## 250          R4   No-till  R2      C         F       10
## 251          R6   No-till  R2      C         F       10
## 252          R3   No-till  R5      A         C       25
```

```
## 253          R4   No-till  R5      A         C       19
## 254          R6   No-till  R5      A         C       17
## 255          R3   No-till  R5      B         C       13
## 256          R4   No-till  R5      B         C       16
## 257          R6   No-till  R5      B         C       19
## 258          R3   No-till  R5      C         C       15
## 259          R4   No-till  R5      C         C        9
## 260          R6   No-till  R5      C         C       19
## 261          R3   No-till  R5      A         F       15
## 262          R4   No-till  R5      A         F        5
## 263          R6   No-till  R5      A         F       12
## 264          R3   No-till  R5      B         F       14
## 265          R4   No-till  R5      B         F        5
## 266          R6   No-till  R5      B         F       17
## 267          R3   No-till  R5      C         F       11
## 268          R4   No-till  R5      C         F        6
## 269          R6   No-till  R5      C         F       28
## 270          R3   No-till  R6      A         C       18
## 271          R4   No-till  R6      A         C       15
## 272          R6   No-till  R6      A         C       19
## 273          R3   No-till  R6      B         C       15
## 274          R4   No-till  R6      B         C       14
## 275          R6   No-till  R6      B         C       20
## 276          R3   No-till  R6      C         C       14
## 277          R4   No-till  R6      C         C       23
## 278          R6   No-till  R6      C         C       24
## 279          R3   No-till  R6      A         F       11
## 280          R4   No-till  R6      A         F        8
## 281          R6   No-till  R6      A         F        7
## 282          R3   No-till  R6      B         F       11
## 283          R4   No-till  R6      B         F        7
## 284          R6   No-till  R6      B         F       14
## 285          R3   No-till  R6      C         F       21
## 286          R4   No-till  R6      C         F        9
## 287          R6   No-till  R6      C         F       14
```

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns
  filter(Treatment == "Conv.") %>% # subsetting to only include the conventional treatment
  mutate(logRich = log(richness)) %>% # creating a new column of the log richness
  head() # displaying the first six rows
```

```
##                                  SampleID Crop Compartment DateSampled
## 1 Corn2017LeafObjective2Collection1T1R1CAH2 Corn        Leaf     6/26/17
## 2 Corn2017LeafObjective2Collection1T1R1CBA3 Corn        Leaf     6/26/17
## 3 Corn2017LeafObjective2Collection1T1R1CCB3 Corn        Leaf     6/26/17
## 4 Corn2017LeafObjective2Collection1T1R1FAC3 Corn        Leaf     6/26/17
## 5 Corn2017LeafObjective2Collection1T1R1FBD3 Corn        Leaf     6/26/17
## 6 Corn2017LeafObjective2Collection1T1R1FCE3 Corn        Leaf     6/26/17
##   GrowthStage Treatment Rep Sample Fungicide richness   logRich
## 1          V6     Conv.  R1      A         C        9 2.1972246
## 2          V6     Conv.  R1      B         C        6 1.7917595
## 3          V6     Conv.  R1      C         C        5 1.6094379
## 4          V6     Conv.  R1      A         F        7 1.9459101
```

```
## 5          V6     Conv.  R1      B         F        4 1.3862944
## 6          V6     Conv.  R1      C         F        2 0.6931472
```

summarise() We can use the `summarise()` function to find things like means and standard deviations/errors.

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns
  filter(Treatment == "Conv.") %>% # subsetting to only include the conventional treatment
  mutate(logRich = log(richness)) %>% # creating a new column of the log richness
  summarise(Mean.rich = mean(logRich)) # calculating overall mean log richness within the conventionall
```

```
##   Mean.rich
## 1  2.304395
```

We can also connect multiple summary statistics here.

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns
  filter(Treatment == "Conv.") %>% # subsetting to only include the conventional treatment
  mutate(logRich = log(richness)) %>% # creating a new column of the log richness
  summarise(Mean.rich = mean(logRich), # calculating the mean richness, stdeviation, and standard error
            n = n(),
            sd.dev = sd(logRich)) %>%
  mutate(std.err = sd.dev/sqrt(n))
```

```
##   Mean.rich   n    sd.dev   std.err
## 1  2.304395 144 0.7024667 0.0585389
```

This is useful, but maybe we want to have summary statistics by group! We can also do this very easily using the `group_by()` function.

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns
  group_by(Treatment, Fungicide) %>% # grouping by treatment and fungicide to later calculate summary s
  mutate(logRich = log(richness)) %>% # creating a new column of the log richness
  summarise(Mean.rich = mean(logRich), # calculating the mean richness, stdeviation, and standard error
            n = n(),
            sd.dev = sd(logRich)) %>%
  mutate(std.err = sd.dev/sqrt(n))
```

```
## `summarise()` has grouped output by 'Treatment'. You can override using the
## `.groups` argument.

## # A tibble: 4 x 6
## # Groups:   Treatment [2]
##   Treatment Fungicide Mean.rich     n sd.dev std.err
##   <chr>     <chr>         <dbl> <int>  <dbl>   <dbl>
## 1 Conv.     C              2.53    72  0.635  0.0748
## 2 Conv.     F              2.07    72  0.696  0.0820
## 3 No-till   C              2.63    72  0.513  0.0604
## 4 No-till   F              2.36    71  0.680  0.0807
```

Connecting to plotting These groups of functions and packages also all work together very nicely. For example, direct input into ggplot is possible. In which case you would not have to enter the data statement, you just need to start inputing your aesthetics.

```
plot1 <- microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns
  group_by(Treatment, Fungicide) %>% # grouping by treatment and fungicide to later calculate summary s
  mutate(logRich = log(richness)) %>% # creating a new column of the log richness
  summarise(Mean.rich = mean(logRich), # calculating the mean richness, stdeviation, and standard error
            n = n(),
            sd.dev = sd(logRich)) %>%
  mutate(std.err = sd.dev/sqrt(n)) %>%
  ggplot(aes(x = Fungicide, y = Mean.rich)) + # adding in a ggplot
  geom_bar(stat="identity") +
  geom_errorbar( aes(x=Fungicide, ymin=Mean.rich-std.err, ymax=Mean.rich+std.err), width=0.4) +
  theme_minimal() +
  xlab("") +
  ylab("Log Richness") +
  facet_wrap(~Treatment)
```

```
## `summarise()` has grouped output by 'Treatment'. You can override using the
## `.groups` argument.
```

Joining The join group of functions are super useful. Have you ever tried to match up data by rows based on some criteria? Its almost impossible without messing up over 10-15 rows of data.

More info can be found here: https://dplyr.tidyverse.org/reference/mutate-joins.html

But we can easily do this with the joining functions. They include:

-left_join() - Keep all rows of X and add matching rows from Y. Any rows in Y that don't match X are excluded. -right_join() - reverse of left_join() -inner_join() - only keep rows that are common to both X AND Y, remove everything else. -full_join() - Keep any columns that are in either X or Y

To try these out lets split our data into separate sets including taxonomy and metadata.

Here we are also sampling 100 random rows of our dataset using the `sample_n()` function

```
# selecting just the richness and sample ID
richness <- microbiome.fungi %>%
  select(SampleID, richness)

# selecting columns that don't include the richness
metadata <- microbiome.fungi %>%
  select(SampleID, Fungicide, Crop, Compartment, GrowthStage, Treatment, Rep, Sample)

head(metadata)
```

```
##                                   SampleID Fungicide Crop Compartment
## 1 Corn2017LeafObjective2Collection1T1R1CAH2         C Corn        Leaf
## 2 Corn2017LeafObjective2Collection1T1R1CBA3         C Corn        Leaf
## 3 Corn2017LeafObjective2Collection1T1R1CCB3         C Corn        Leaf
## 4 Corn2017LeafObjective2Collection1T1R1FAC3         F Corn        Leaf
## 5 Corn2017LeafObjective2Collection1T1R1FBD3         F Corn        Leaf
## 6 Corn2017LeafObjective2Collection1T1R1FCE3         F Corn        Leaf
```

```
##   GrowthStage Treatment Rep Sample
## 1          V6     Conv.  R1      A
## 2          V6     Conv.  R1      B
## 3          V6     Conv.  R1      C
## 4          V6     Conv.  R1      A
## 5          V6     Conv.  R1      B
## 6          V6     Conv.  R1      C
```

```
head(richness)
```

```
##                                   SampleID richness
## 1 Corn2017LeafObjective2Collection1T1R1CAH2        9
## 2 Corn2017LeafObjective2Collection1T1R1CBA3        6
## 3 Corn2017LeafObjective2Collection1T1R1CCB3        5
## 4 Corn2017LeafObjective2Collection1T1R1FAC3        7
## 5 Corn2017LeafObjective2Collection1T1R1FBD3        4
## 6 Corn2017LeafObjective2Collection1T1R1FCE3        2
```

```
head(left_join(metadata, richness, by = "SampleID")) # adding the richness data to the metadata based o
```

```
##                                   SampleID Fungicide Crop Compartment
## 1 Corn2017LeafObjective2Collection1T1R1CAH2         C Corn        Leaf
## 2 Corn2017LeafObjective2Collection1T1R1CBA3         C Corn        Leaf
## 3 Corn2017LeafObjective2Collection1T1R1CCB3         C Corn        Leaf
## 4 Corn2017LeafObjective2Collection1T1R1FAC3         F Corn        Leaf
## 5 Corn2017LeafObjective2Collection1T1R1FBD3         F Corn        Leaf
## 6 Corn2017LeafObjective2Collection1T1R1FCE3         F Corn        Leaf
##   GrowthStage Treatment Rep Sample richness
## 1          V6     Conv.  R1      A        9
## 2          V6     Conv.  R1      B        6
## 3          V6     Conv.  R1      C        5
## 4          V6     Conv.  R1      A        7
## 5          V6     Conv.  R1      B        4
## 6          V6     Conv.  R1      C        2
```

Now this is kind of a silly example because we already had the data set joined how we wanted it and split it apart. But you can imagine if you have two separate data sets with one column in common and you wanted to combine them into one, instead of taking days to do this by hand you could simply load into R and do it.

Pivoting Pivoting is also useful for converting from wide to long format and back again. We can do this with `pivot_longer()` and `pivot_wider()`

More info can be found here: https://tidyr.tidyverse.org/reference/pivot_wider.html

I will demonstrate `pivot_wider()` assuming that most data is already formatted as long data. Lets explore turning our data set into a wide format, since its in long format.

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>%
  group_by(Treatment, Fungicide) %>% # grouping by treatment and fungicide to later calculate summary s
  summarise(Mean = mean(richness)) # calculates the mean per Treatment and Fungicide
```

```
## `summarise()` has grouped output by 'Treatment'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 4 x 3
## # Groups:   Treatment [2]
##   Treatment Fungicide  Mean
##   <chr>     <chr>     <dbl>
## 1 Conv.     C         14.6
## 2 Conv.     F          9.75
## 3 No-till   C         15.4
## 4 No-till   F         13.1
```

Wide format - sets the values within the fungicide column into column names

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns
  group_by(Treatment, Fungicide) %>% # grouping by treatment and fungicide to later calculate summary s
  summarise(Mean = mean(richness)) %>% # calculates the mean per Treatment and Fungicide
  pivot_wider(names_from = Fungicide, values_from = Mean) # pivot to wide format
```

```
## `summarise()` has grouped output by 'Treatment'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 2 x 3
## # Groups:   Treatment [2]
##   Treatment     C     F
##   <chr>     <dbl> <dbl>
## 1 Conv.      14.6  9.75
## 2 No-till    15.4 13.1
```

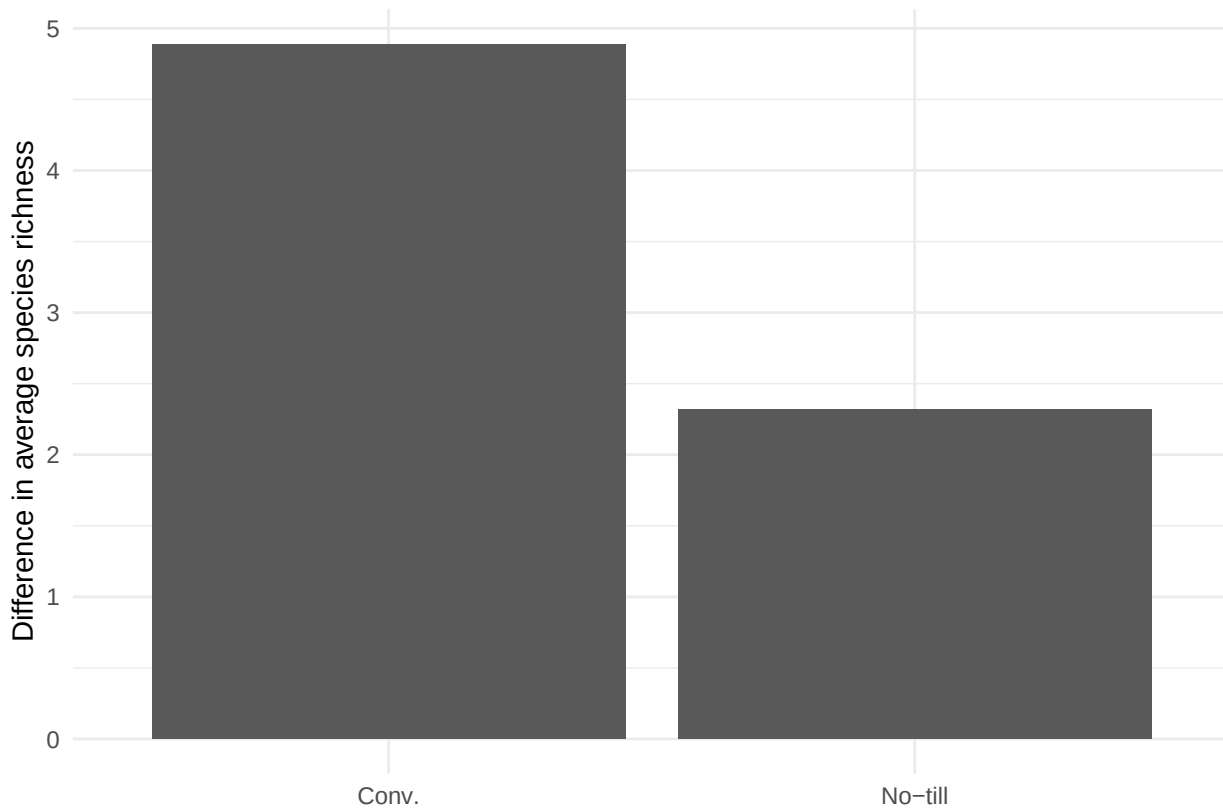
Easily can take the difference bewteen the fungicide and control now.

```
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns  filter(Class == "Sor
  group_by(Treatment, Fungicide) %>% # grouping by treatment and fungicide to later calculate summary s
  summarise(Mean = mean(richness)) %>% # calculates the mean per Treatment and Fungicide
  pivot_wider(names_from = Fungicide, values_from = Mean) %>% # pivot to wide format
  mutate(diff.fungicide = C - F) # calculate the difference between the means.
```

```
## `summarise()` has grouped output by 'Treatment'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 2 x 4
## # Groups:   Treatment [2]
##   Treatment     C     F diff.fungicide
##   <chr>     <dbl> <dbl>          <dbl>
## 1 Conv.      14.6  9.75           4.89
## 2 No-till    15.4 13.1            2.32
```

Now we can easily calculate the difference between fungicide and control and plot it.

```r
microbiome.fungi %>%
  select(SampleID, Crop, Compartment:Fungicide, richness) %>% # selecting columns  filter(Class == "Sor
  group_by(Treatment, Fungicide) %>% # grouping by treatment and fungicide to later calculate summary s
  summarise(Mean = mean(richness)) %>% # calculates the mean per Treatment and Fungicide
  pivot_wider(names_from = Fungicide, values_from = Mean) %>% # pivot to wide format
  mutate(diff.fungicide = C - F) %>%  # calculate the difference between the means.
  ggplot(aes(x = Treatment, y = diff.fungicide)) + # Plot it
  geom_col() +
  theme_minimal() +
  xlab("") +
  ylab("Difference in average species richness")
```

```
## `summarise()` has grouped output by 'Treatment'. You can override using the
## `.groups` argument.
```